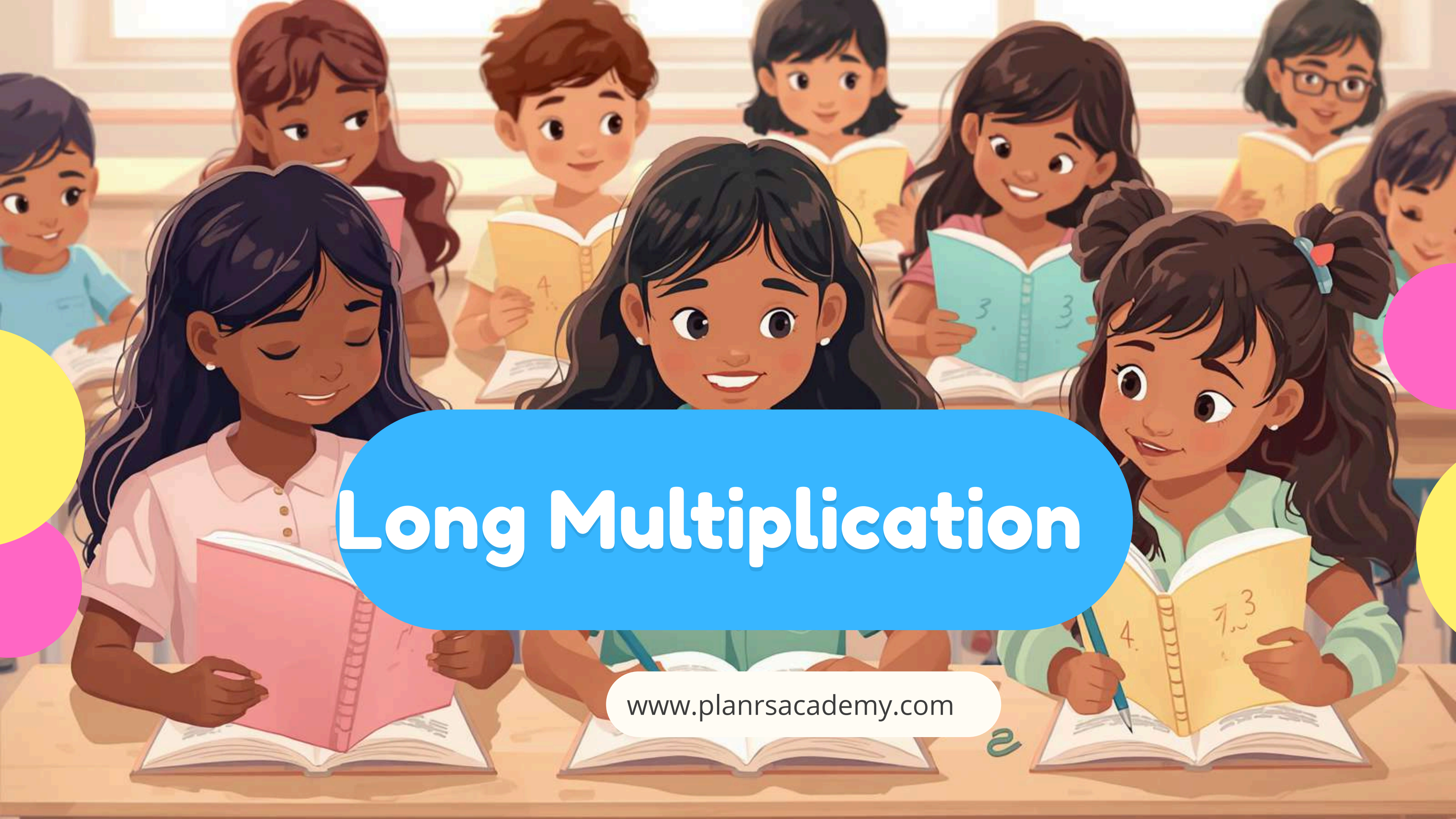


A vibrant illustration of a diverse group of young students in a classroom. In the foreground, three girls are seated at a wooden desk, each with an open book. The girl on the left, with long dark hair and a pink shirt, is looking down at her pink book. The girl in the center, with long dark hair and a green shirt, is looking up and smiling. The girl on the right, with dark hair in pigtails and a green shirt, is looking at her yellow book which shows a long multiplication problem:
$$\begin{array}{r} 1.3 \\ \times 4 \\ \hline \end{array}$$
. Other students are visible in the background, also reading books. The scene is bright and cheerful, with large yellow and pink circles on the left and right sides of the image.

Long Multiplication

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What is Long Multiplication?

Long multiplication is a special method that helps us multiply bigger numbers easily! When we need to multiply two 2-digit numbers like 23×15 , it can seem tricky. But with long multiplication, we break it down into simple steps. We multiply one digit at a time, write down partial products, and then add them together. It's like solving a puzzle piece by piece!



Understanding Place Values: Tens and Ones



Ones Place:
The units digit tells us how many single items. In 34, the digit 4 is in the ones place.

Tens Place:
The tens digit tells us how many groups of 10. In 34, the digit 3 means 30 (three tens).



Breaking Down Numbers:
 $34 = 30 + 4$
Tens + Ones = Total!

Step 1

Multiply Ones Digit



Example: 12×13

Look at the ones digit of 13, which is 3.

- First: 3×2 (ones of 12) = 6 ✓
 - Then: 3×1 (tens of 12) = 3 ✓
 - Write the partial product: 36
- This is our first partial product!

Long Multiplication Example

Forgetting Numbers Pains

$$19 = 12 \Rightarrow 12$$

$$13 = 13 \Rightarrow 15$$

$$14 = 13 \Rightarrow 15$$

$$14 = 13 \Rightarrow 14$$

$$14 = 13 \Rightarrow 15$$

$$12 = 13 \Rightarrow 14$$

The Rule:

Multiply the ones digit of the second number by each digit of the first number. Start from the right side! If the product is greater than 9, carry over the tens digit to the next column.



Step 2 Multiply Tens Digit



Remember:

When multiplying by the tens digit, always add a zero at the end! This is because the tens digit represents 10 times its value. If your multiplication gives a number bigger than 9, carry over the extra digit to the next column, just like in Step 1.

Long Multiplication
Step # 2

$$\begin{array}{r} 12 \\ \times 13 \\ \hline \end{array}$$

$1 = 1 \times 2 = 2$

$18 + 10 = 28$

28

How It Works:

Now multiply the tens digit of the second number (1 in 13) by both digits of the first number (12). Remember: Add a zero as a placeholder because we're multiplying by tens!

$1 \times 2 = 2$, then $1 \times 1 = 1$

Write 120 (with zero placeholder)



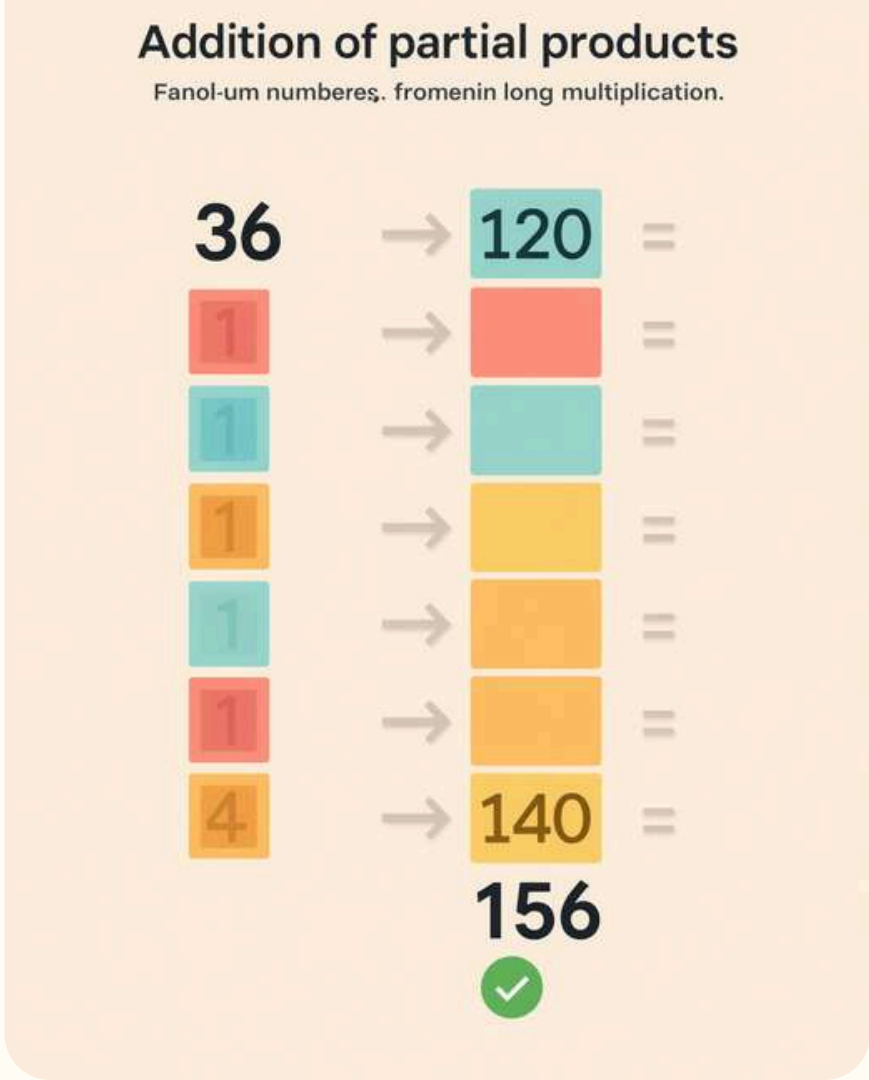
Step 3

Add Partial Products



Example: 12 × 13

From Step 1: $12 \times 3 = 36$
From Step 2: $12 \times 10 = 120$
Now add them:
36
• 120
156
So, $12 \times 13 = 156!$ ✓



How to Add:

Line up the partial products by place value. The ones digit goes under ones, tens under tens. Then add column by column from right to left. This gives you the final answer!



Visual Representation with Example

Let's solve a real problem! If one mango costs ₹34, how much do 26 mangoes cost?

Step 1: Multiply 34×6 (ones) = 204

Step 2: Multiply 34×20 (tens) = 680

Step 3: Add partial products: $204 + 680 = ₹884$

So, 26 mangoes cost ₹884!

Oplent-nallyi prtuttsrpliation




$$34 \times 26 = 26$$

$$34 = 26 +$$

₹	204	+24	26	+261
₹ =	30	+31	+6	++6

₹	= ₹	280	+30	+3
₹ =	₹	₹	881	+348

Practice Problems with Solutions



Problem 1: $23 \times 15 = 345$

Step: $(23 \times 5 = 115) +$
 $(23 \times 10 = 230)$

Problem 2: $45 \times 12 = 540$

Step: $(45 \times 2 = 90) +$
 $(45 \times 10 = 450)$

Problem 3: $52 \times 24 = 1248$

Step: $(52 \times 4 = 208) +$
 $(52 \times 20 = 1040)$

Try solving before checking answers!

16



Real-World Applications



Counting in Groups:
How many pencils in 24
boxes with 12 pencils
each? Multiply to find out!

Calculating Costs:
If one notebook costs
₹35, how much for 15
notebooks? Use long
multiplication!



Measuring Distances:
If a cricket pitch is 22
meters, what's the length
of 14 pitches combined?



Common Mistakes to Avoid



Forgetting Zero Placeholder:
Always add a zero when
multiplying by the tens digit!

Misaligning Partial Products:
Line up numbers by place
value - ones under ones,
tens under tens.

Forgetting to Carry Over:
When a product is 10 or
more, carry the tens digit to
the next column.



Skipping Steps or Rushing:
Take your time! Check each
step carefully before moving
on.



Tips and Tricks for Easier Calculation



$$= 2 - 4 =$$



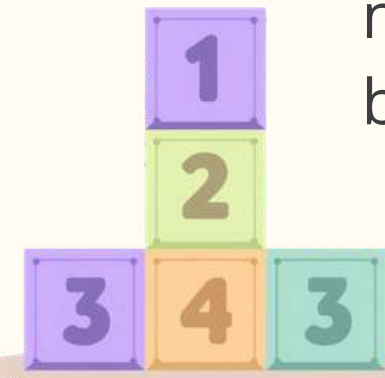
Write Clearly: Always write numbers neatly and align them properly by place value.



Double-Check Steps: Verify each multiplication step before moving to the next one.



Use Estimation: Round numbers to estimate answers and check if your final answer makes sense.



Start Small: Practice regularly with smaller numbers first, then try bigger ones!

Worked Example 1



Hare by star is ternbis products?

$12 + 1$
 $369 = + -$

$2 + 13$
 $120 + = 3$

$120 = + =$
 $100 = - -$

$120 = + -$
 $120 = - -$

$120 + + = + 105 = 166$

$30 + + = + 145 = 156$

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Easy Level:

Let's solve 12×13 step by step!
Step 1: Multiply by ones (3)
 $12 \times 3 = 36$ (first partial product)
Step 2: Multiply by tens (1)
 $12 \times 10 = 120$ (second partial product)
Step 3: Add partial products
 $36 + 120 = 156$
Answer: $12 \times 13 = 156$

Real-Life Example:

Imagine you want to buy pens for your class!
If one pen costs ₹12 and you need 13 pens for your friends, how much will you spend?
Using long multiplication:
 $₹12 \times 13 = ₹156$
You will spend ₹156 on pens! ✍️



Worked Example 2



Real-Life Problem:

At the school canteen, one packet of chips costs ₹34. If you want to buy 26 packets for your class party, how much will you spend?
Answer: $34 \times 26 = ₹884$
You will spend ₹884 on chips for the party!

8844 olulc multiplication

Dece ppojer multieing outh attacked for products

3

+

4

✖

50

=

36

1

+

2

✖

304

=

43

2

+

3

✖

80d#

=

36

4

+

20

✖

25*

=

76

3

+

246

✖

90*

=

26

5

+

848

✖

358

=

25

8

+

8250

✖

186*

=

842

5

+

260

✖

698

=

1180+

www.treaseshdtesonmre.com

TABLE 84+

34 × 26:

Step 1: Multiply 34 by 6 (ones digit)
 $34 \times 6 = 204$ (carry the 2, then $3 \times 6 + 2 = 20$)
Step 2: Multiply 34 by 20 (tens digit)
 $34 \times 2 = 68$, add placeholder zero = 680
Step 3: Add partial products
 $204 + 680 = 884$



Worked Example 3

Challenging Level



Solution:

87

× 95

435 ← (87 × 5)

7830 ← (87 × 90)

8265 ✓

The train travels 8,265 kilometers!

Remember: Carry over carefully when products exceed 9.

Long multiplication

Carrying over data ranid out soniprocires, Plarce 6 tudes T+ 2,96+ 72,75,

87 = 95 95

Carrying over digits

Paret mutii producti

Ollying value milx 7-830 Clowcarrying over digits: s fin 15113.

435 = 78 0

Carrying over digits

420 3 = 78(0

4.(= 78(0

Carrying over digits

145 = 48 4436

435 = 45 4418

Bew n past doug

Problem:

Let's solve 87 × 95 step by step!

Context: A train travels 87 km per hour. How far does it travel in 95 hours?

Step 1: Multiply by ones (5)

87 × 5 = 435

Step 2: Multiply by tens (9) with placeholder

87 × 90 = 7830

Step 3: Add partial products

435 + 7830 = 8,265 km



Your Turn to Try!



Now it's time to practice what you've learned! Solve this problem step by step:

$$56 \times 27 = ?$$

Remember:

- First, multiply 56 by 7 (ones digit)
- Then, multiply 56 by 20 (tens digit with zero placeholder)
- Finally, add your partial products

Write down each step in your notebook and check your answer!



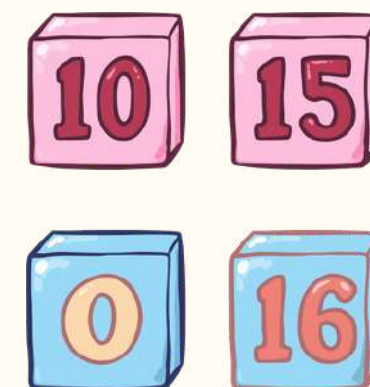
Summary and Key Takeaways



MATH: _____

 **MULTIPLICATION**
This the matel accaptors of the idey the arp woestiprations

$3 \times 0 =$ ☒	☒ $\times 16$	☒ $\times 5 =$ ☐
$3 \times 2 =$ ☒	☒ $\times 12$	☒ $\times 3 =$ ☐
$3 \times 5 =$ ☒	☒ $\times 12$	☒ $\times 6 =$ ☐
$2 \times 5 =$ ☒	☒ $\times 15$	☒ $\times 8 =$ ☐
$7 \times 2 =$ ☒	☒ $\times 15$	☒ $\times 4 =$ ☐
$2 \times 4 =$ ☒	☒ $\times 15$	☒ $\times 3 =$ ☐
$2 \times 9 =$ ☒	☒ $\times 13$	☒ $\times 5 =$ ☐
$2 \times 5 =$ ☒	☒ $\times 12$	☒ $\times 4 =$ ☐
$3 \times 9 =$ ☒	☒ $\times 16$	☒ $\times 2 =$ ☐
$7 \times 0 =$ ☒	☒ $\times 15$	☒ $\times 6 =$ ☐
$3 \times 9 =$ ☒	☒ $\times 16$	☒ $\times 5 =$ ☐



Understand Place Values:
Know your tens and ones
before multiplying any
numbers.

Multiply Step-by-Step: First
multiply by ones digit, then
tens digit with zero
placeholder.

Add Partial Products: Line
up numbers by place value
and add carefully.

Practice Makes Perfect:
Regular practice with
different problems builds
confidence and speed!

Thank You!

Great job learning long multiplication! Keep practicing and you will master it. Remember, practice makes perfect!



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